Obsidian RAG Model

This is your space for legal copy.
Mention if the information is confidential and who should to read this deck.

Agenda

1) Project Overview

2) Objectives

3) Key features

4) Process

5) Questions

What’s Obsidian?

Obsidian

Obsidian is one of the most popular free note-taking app with plugin support, with 100 million total downloads

It prides itself on being lightweight, with all data staying private and local to the user.

Context by Consent

COMPLETE PRIVACY
Locally Hosted LLM

TRANSPARENCY
Top K Retrieval

CUSTOMIZABILITY
Programmable Context

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Goals & Outcomes

LEARNING REINFORCEMENT

1. Local RAG System Design

   Hardware constraints, latency vs quality optimization

2. Embedding & Context Management

   Chunking, File-level Embedding, Semantic Abstraction

3. Event Driven ML Pipelines

   Event-based Ingestion, Debouncing

PROJECT GOALS

1. Demo MVP at a hackathon
2. Technical project for resume flexing
3. Ethical tool that I might actually use

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MVP Capabilities/Features

1

Local LLM Inference

A stand-alone locally hosted model, without any reliance on the web or LLM APIs. Complete privacy with predictable behavior.

2

Explicit Control over Context

Instead of hidden and automatic context management, users can directly edit the context by white or black-listing files, folders, or tags.

3

Source Transparency

Response references notes directly. User can verify responses as it generates in real time.

4

User Curated Context

Everything the model can see will be inside a personal vault. All the knowledge reinforcing the model will come from the user themselves.

5

Web-based Retrieval

A bonus opt-in feature if just the vault's notes feels inadequate for quality responses.

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Obsidian Plugin Ecosystem

1 Excellent Plugin Support

Well maintained developer guides and many open source plugins for AI to munch on

2 Local Architecture

All notes and context are stored locally in a simple file structure

3 Pre-packaged UI & Integration

Simple to deploy to the Obsidian community plugin marketplace

Basic User Flow

1

Step one

User write edits their vault notes, a file watcher listens for edit, and parses the edited text into metadata/tags

2

Step two

User queries their locally hosted model, and the model outputs an initial response by pulling files based on metadata

3

Step three

If the user is unhappy with the response, they can directly edit the context by white or black listing notes

Proposed Tech Stack

Frontend

React + Typescirpt

File Watching

chokidar

Local LLM Runtime

Ollama

Embedding Model

Dunno yet, prob nomic

Storage

SQLite

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Ask me anything!

durpdur

Target audiences

Small Businesses

- Internal documentation, and project notes as context
- On-device AI for sensitive or proprietary information
- No reliance on cloud services or external data sharing

Students

- Reliant responses
- Self curated notes